

Missouri University of Science and Technology

Intro to Data Science
CS 3402 (Spring 2026)

Lecture 2 Introduction to DS

Xiaowei Yu, Ph.D.

Department of Computer Science

Notes

We need volunteers to present Python/Timm/PyTorch/IDEs. (send me you name and MST ID)

Will upload the project/homework to Canvas.

TA: Jiaqian Zhu (jzbmn@mst.edu)

PyTorch & IDE: Kaiser, Luke (ltkhg4@mst.edu) Chenevert, William (wucmxv@mst.edu) (Jan. 29th)

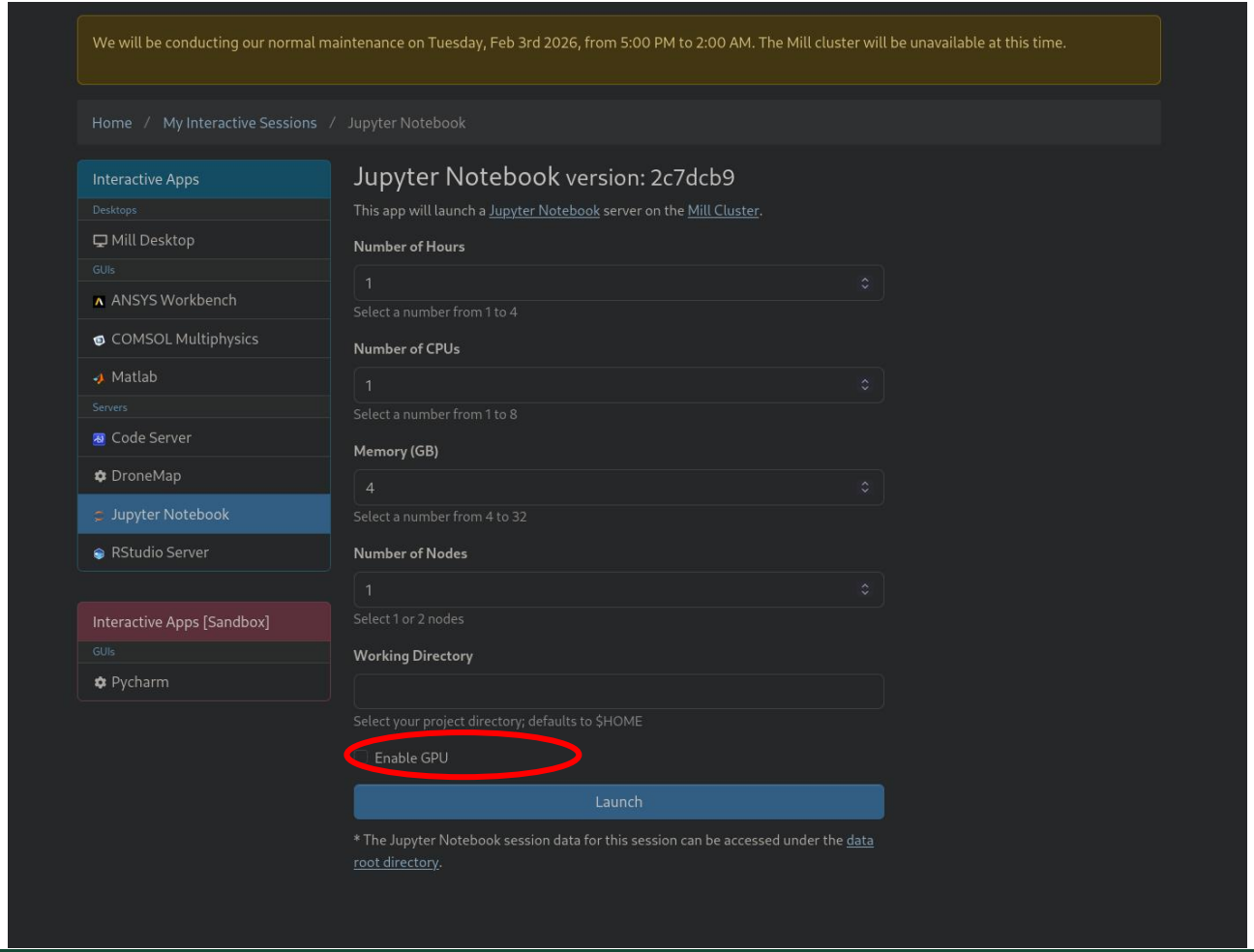
Python & Timm: (Jan. 27th)

GPU nodes

GPU Clusters

To use the GPUs in those nodes, students simply need to click the **“Enable GPU”** checkbox at the bottom of the resource request form

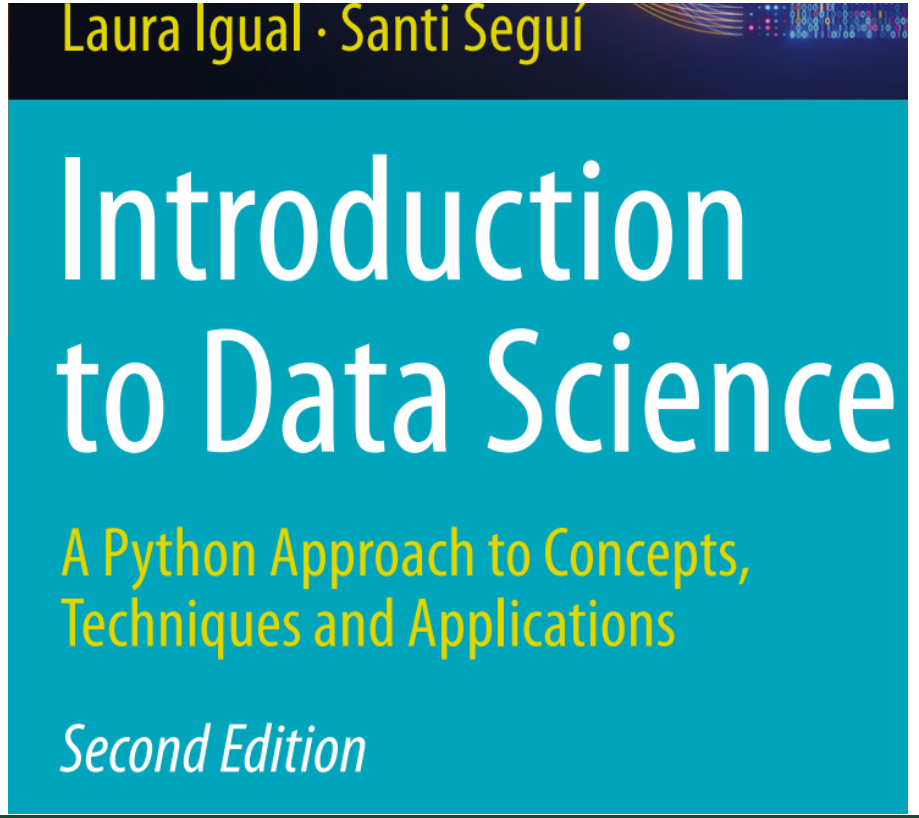
This option is only available for **Jupyter Notebook**; but enabling it for the other **Interactive Apps** is quite simple, you need only ask and we can get it added to whichever you need.



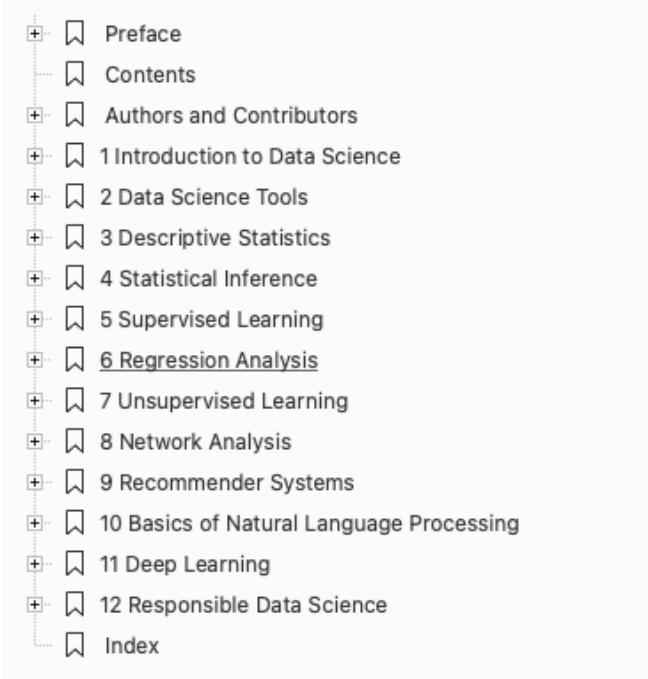
Introduction to Data Science

Optional, Not mandatory

Free PDF is available via University Library



Introduction to Data Science



Data Science Tools

Supervised Learning

Unsupervised Learning

Regression

NLP

Deep Learning

Introduction to Data Science

Python

Pytorch

IDE

Jan 2026	Jan 2025	Change	Programming Language	Ratings	Change
1	1		Python	22.61%	-0.68%
2	4	↑	C	10.99%	+2.13%
3	3		Java	8.71%	-1.44%
4	2	↓	C++	8.67%	-1.62%
5	5		C#	7.39%	+2.94%
6	6		JavaScript	3.03%	-1.17%
7	9	↑	Visual Basic	2.41%	+0.04%
8	8		SQL	2.27%	-0.14%
9	11	↑	Delphi/Object Pascal	1.98%	+0.19%
10	18	⇈	R	1.82%	+0.81%
11	32	⇈	Perl	1.63%	+1.14%
12	10	↓	Fortran	1.61%	-0.42%
13	14	↑	Rust	1.51%	+0.34%
14	15	↑	MATLAB	1.40%	+0.34%

Introduction to Data Science





















What is Data Science?

Solving problems with data



What makes a good data scientist?



What is Data Science?

Solving problems with data



Which step is most challenging?

Use data to create solution

- Data analysis
- Machine learning
- Deep learning

Explore the world using data

Strategies to explore the world using data

- Probing Reality
  - Passive methods: Statistical and machine learning be used in stable environments
  - Active methods: Taking decisions about subsequent actions.
- Pattern Discovery
- Prediction future events
- Understanding human perception: Scientific understanding of cognitive abilities, e.g, CV, NLP, Speech processing

What is Data Analysis?

Using data to discover useful information



- Data: anything you can measure or record



- Statistics: summarize (and visualize) main characteristics of the data

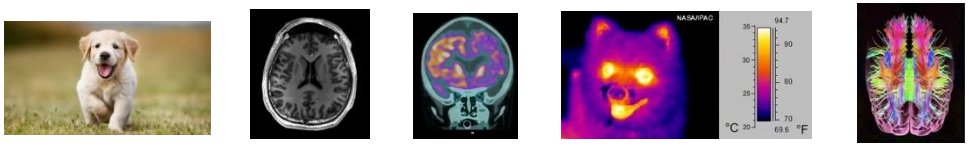


- Algorithms: apply algorithms to find patterns in the data

Data



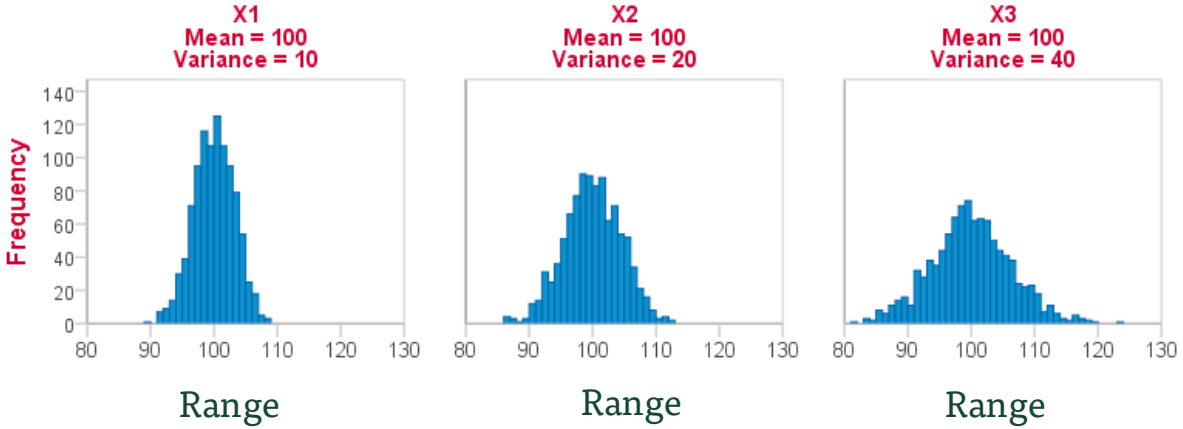
- Data: anything you can measure or record

		Label
Text data	Wall St. Bears Claw Back Into the Black (Reuters) Reuters - Short-sellers, Wall Street's dwindling\band of ultra-cynics, are seeing green again.	Business
Image data		
Video data		
Graph data	 Social networks	

Statistics



- Statistics: summarize (and visualize) main characteristics of the data



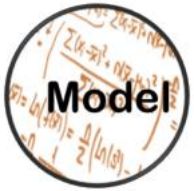
Mean, Median, STD, Range, Variance, etc..

What is Machine Learning?

Create and use models that learn from data



- Data: anything you can measure or record



- Model: specification of a (mathematical) relationship between different variables



- Evaluation: how well does the model work?

What is Machine Learning?

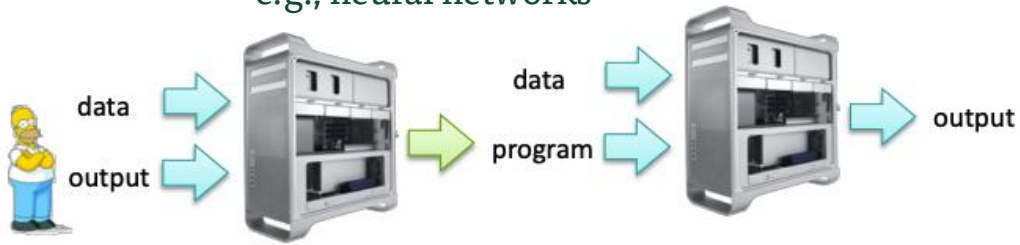
- Traditional CS

e.g., signal processing algorithms, FFT



- Machine Learning

e.g., neural networks

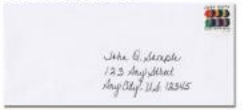


What is Machine Learning?

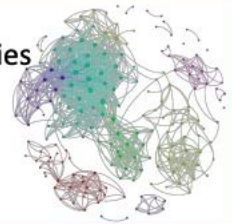
Create and use models that learn from data

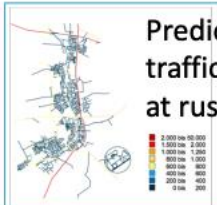
Examples

Identifying zip code from handwritten digits



Detecting communities in social networks





Predicting the traffic volume at rush hour

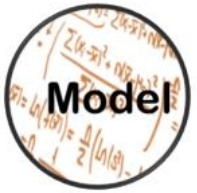
Detecting fraudulent credit card transactions



Determining the location of distribution centers based on customers' residence



Learning from Data

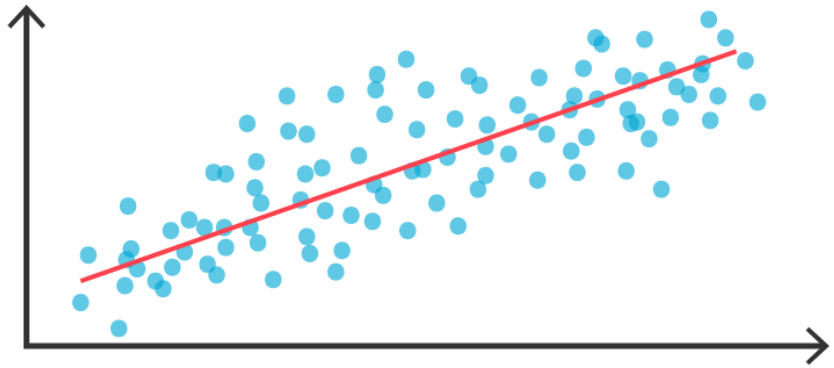


Regression

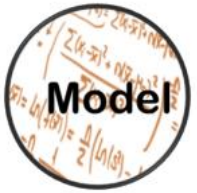
Stock Market Prediction



Data Fitting



Learning from Data

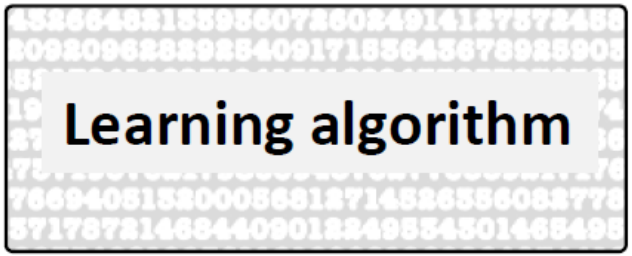


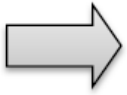
Regression

Training data

$\{(X_i, Y_i)\}_{i=1}^{n}$



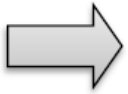


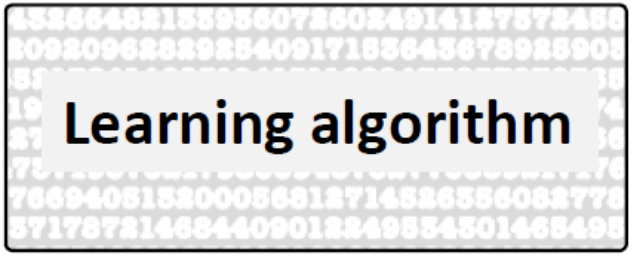


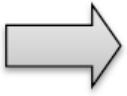
Prediction rule

$\widehat{f}_n$

Testing data

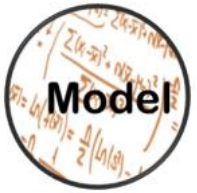




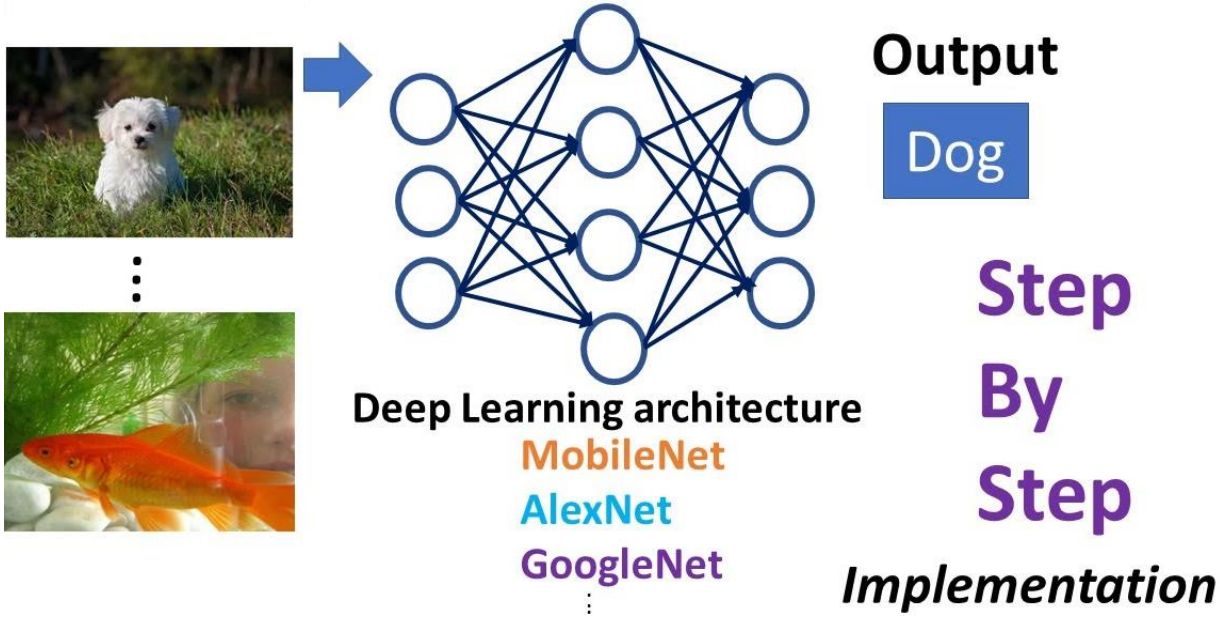


Predicted results

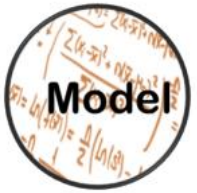
Learning from Data



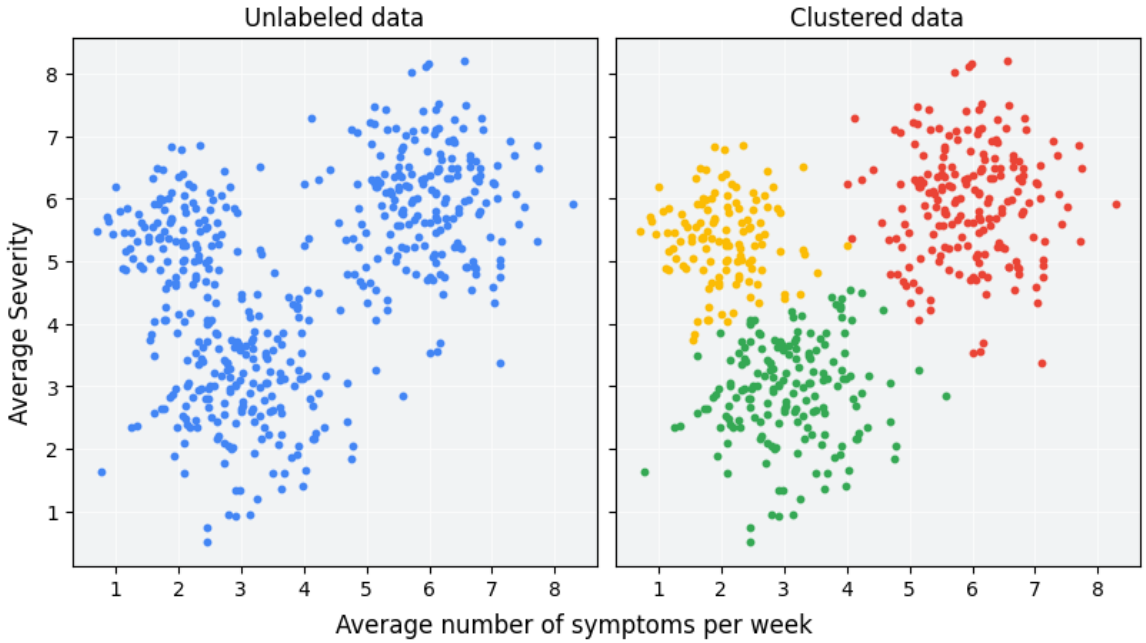
Classification



Learning from Data



Clustering



Unsupervised learning

What is Machine Learning?

Create and use models that learn from data

→ Regression, classification, clustering.

1. Come up with predictions → Supervised learning/predictive modelling
2. Extract knowledge/insights → Unsupervised learning/data mining

Activity

Create and use models that learn from data

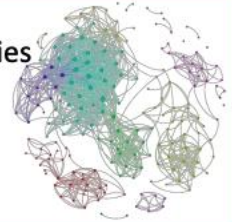
→ Regression, classification, clustering.

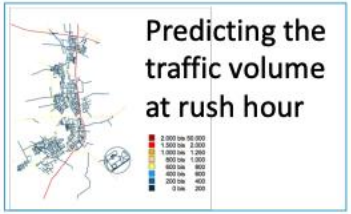
Categorize these Examples

Identifying zip code from handwritten digits



Detecting communities in social networks





Detecting fraudulent credit card transactions



Determining the location of distribution centers based on customers' residence

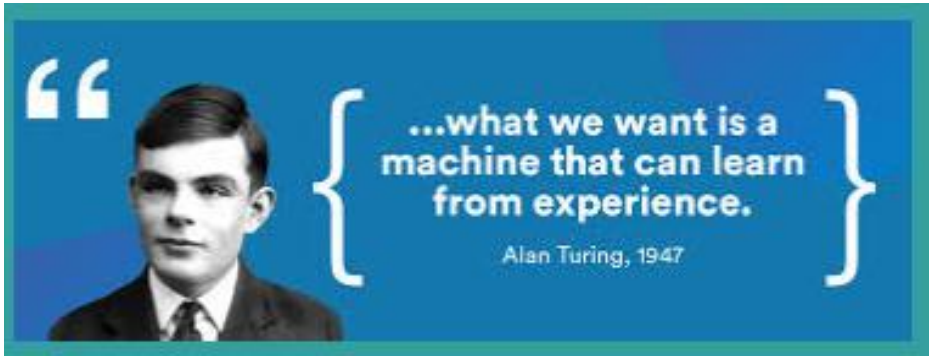


Summary & Reading

Data Science is about

Data, **models**, and **evaluation**

Data Science can solve a wide variety of problems – once we have the right **data** and **model**



Data Science with AI

Visual Object Categorization



We are given categories for these images:

From ETH database of object categories, [Leibe & Schiele 2003]

A classification problem: predict category y based on image x.

Little chance to "hand-craft" a solution, without learning.

Applications: robotics, HCI, web search (a real image Google…)



What are these?

Data Science with AI

Object Detection





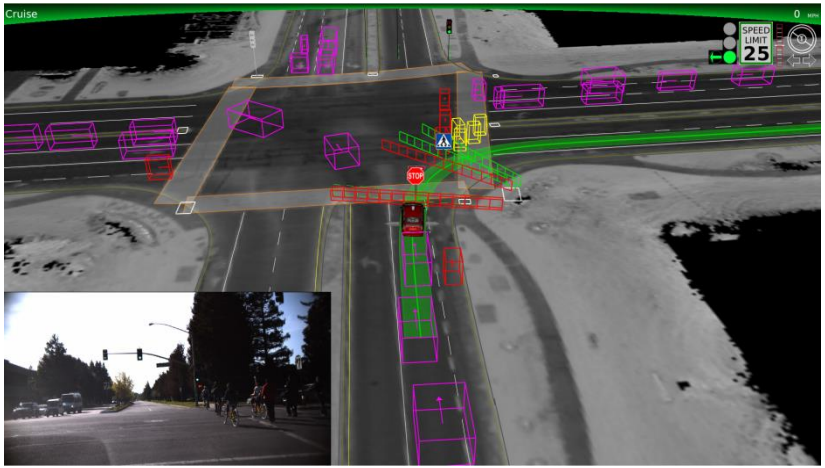
Example training images for each orientation (Prof. H. Schneiderman)



Data Science with AI

Self-driving





Data Science with AI

Text Classification



Company home page

vs

Personal home page

vs

University home page

vs

...

More Applications

There are already a number of applications

- face, speech, handwritten character recognition
- fraud detection (e.g., credit card)
- recommender problems (e.g., which movies/products/etc you'd like)
- annotation of biological sequences, molecules, or assays
- market prediction (e.g., stock/house prices)
- finding errors in computer programs, computer security
- defense applications
- etc

Fundamental Python Libraries for Data Scientists

Numpy		**Python library used for working with arrays**
Pandas		**Pandas offers data structures and operations for manipulating numerical tables and time series**
Matplotlib		**Static, interactive, and animated visualizations.**
Scikit-learn		**Machine learning library for the Python programming language.**

Fundamental Libraries for Machine Learning

The most popular library (PyTorch) for AI model development in both industry and academia.



Useful Resources



Hugging Face

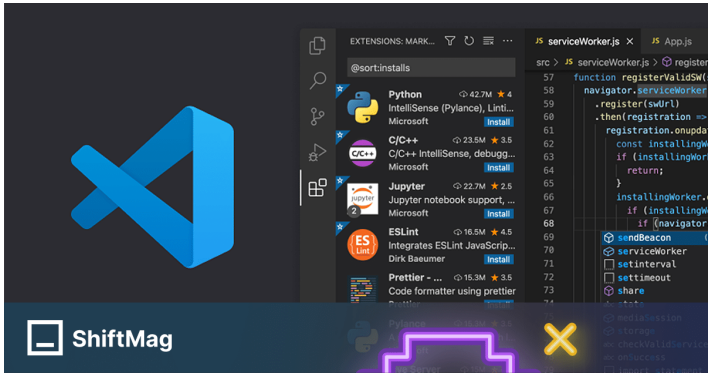
Popular platform for the machine learning community
Especially for NLP research.



GitHub

Developer platform that allows developers to create, store, manage, and share their code.

Integrated development environment (IDE)



VSCode

PyCharm

jupyter

Homework

Install Python, Numpy, Matplotlib on your computer.

Install a IDE (VSCode/Pycharm/Jupyter Notebook) on your computer.

Preview

01/27: Data Science Tools – Python/Libraries

01/29: Data Science Tools – IDE and PyTorch